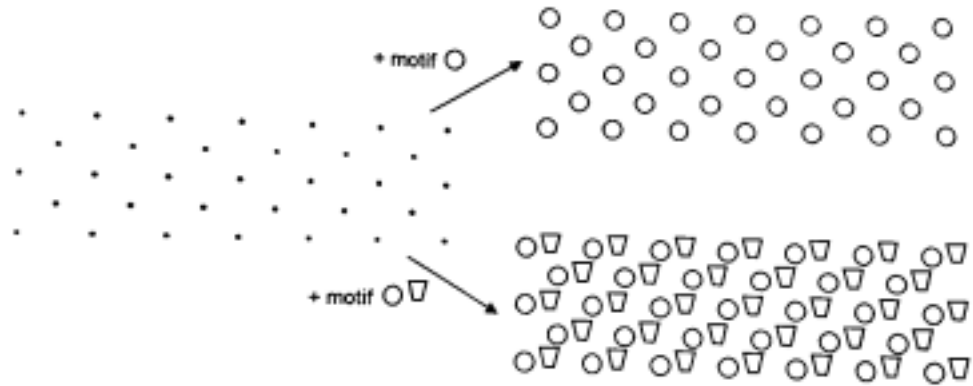
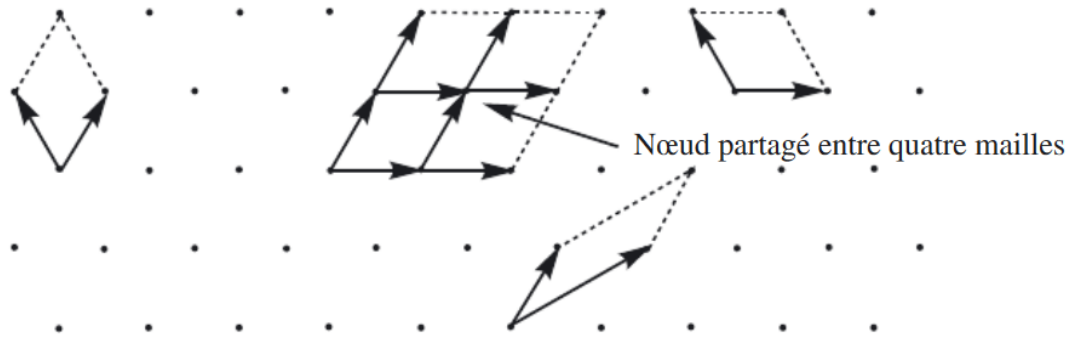


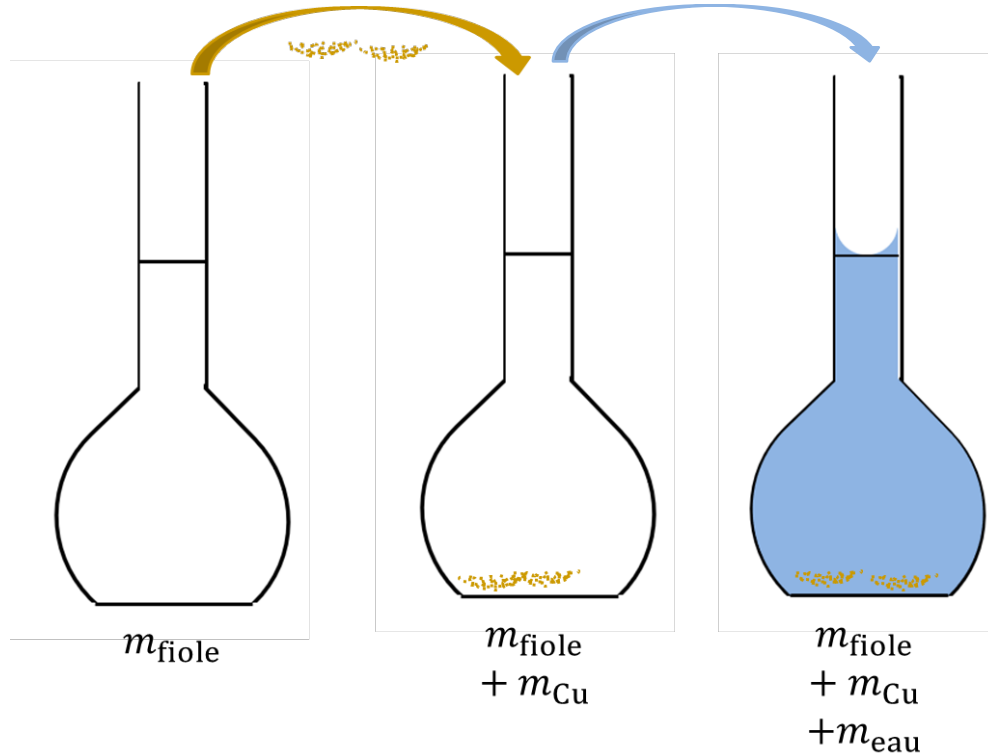
Solides cristallins

Agrégation

Réseaux et Motifs



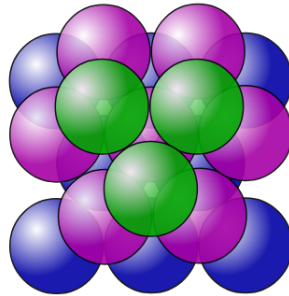
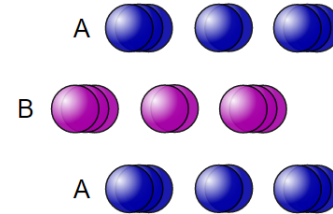
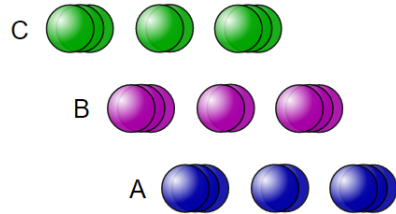
Mesure de la masse volumique du Cuivre



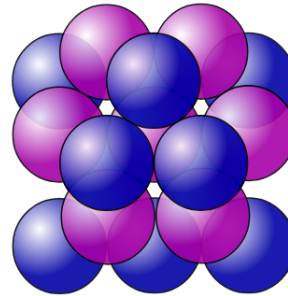
$$V_{\text{Cu}} = V_{\text{fiolle}} - \frac{m_{\text{eau}}}{\rho_{\text{eau}}}$$

$$\rho_{\text{Cu}} = \frac{m_{\text{Cu}}}{V_{\text{Cu}}}$$

Empilement compacts

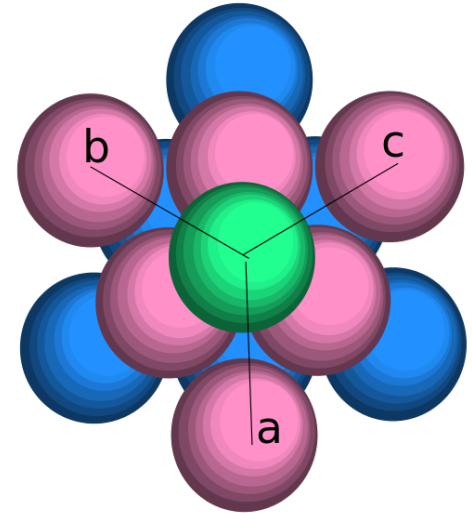
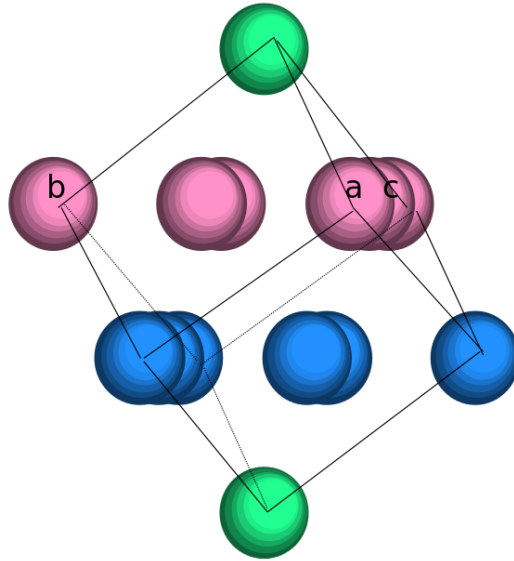
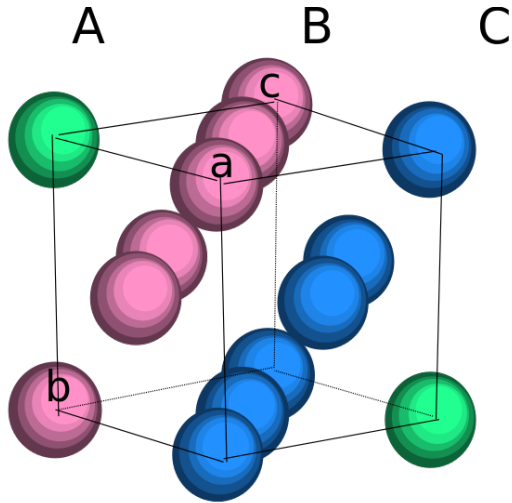


Cubique à faces centrées ABC

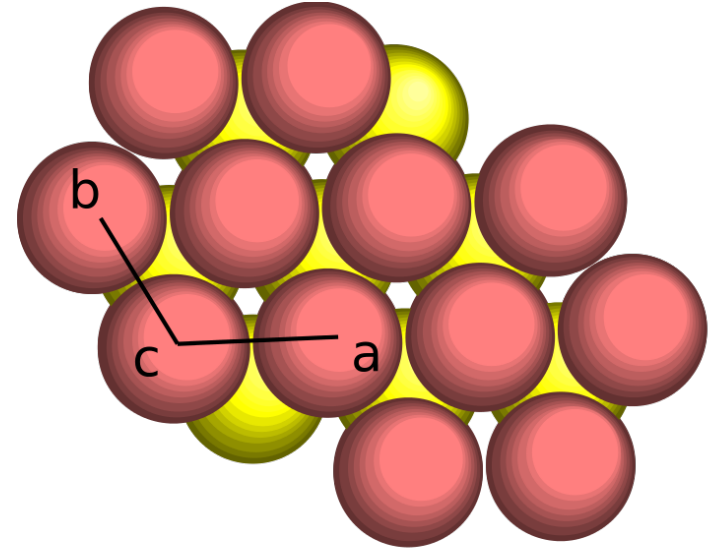
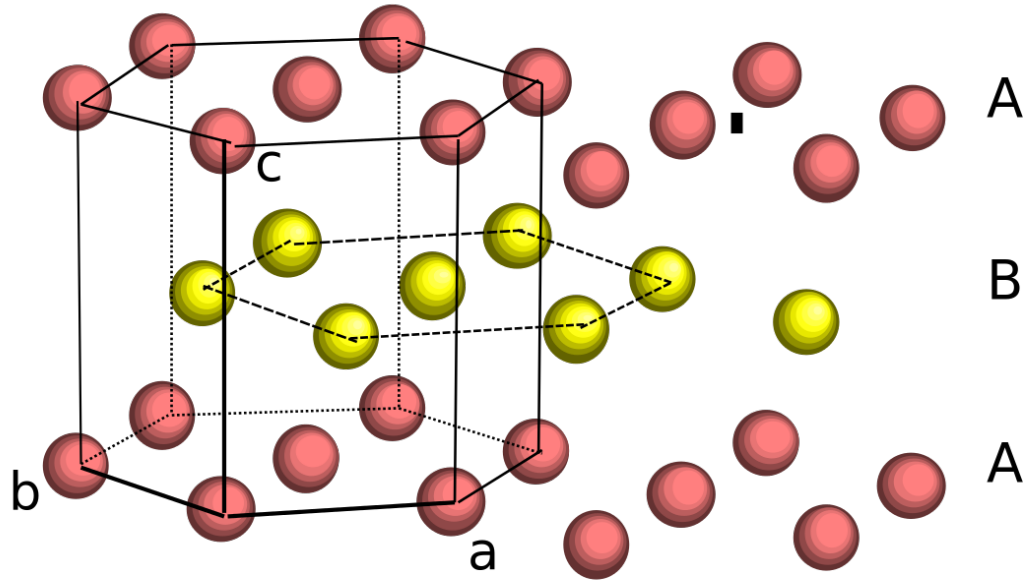


Hexagonal compact ABA

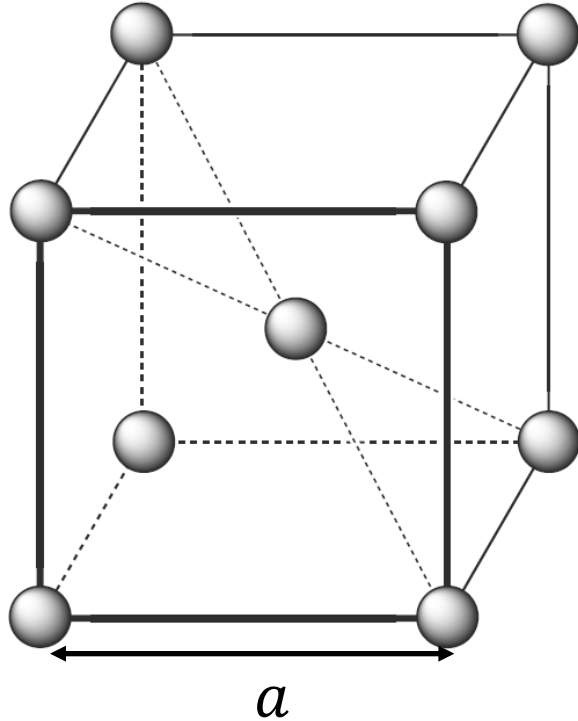
Structure cubique face centrées



Structure hexagonale compacte



Structure cubique centré



- Condition de tangence :

$$a\sqrt{3} = 4r \Rightarrow \boxed{a = \frac{4r}{\sqrt{3}}}$$

- Population :

$$8 \times \frac{1}{8} + 1 = \boxed{2 \text{ motifs par maille}}$$

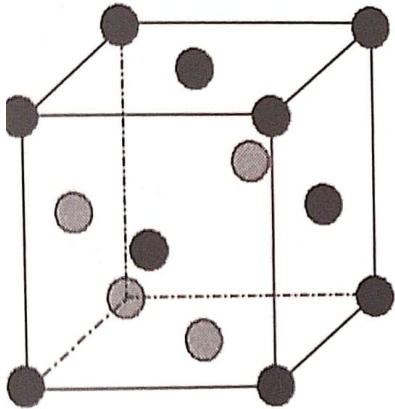
- Coordinnence :

$$\boxed{\text{Fe/Fe} = [8]}$$

- Compacité :

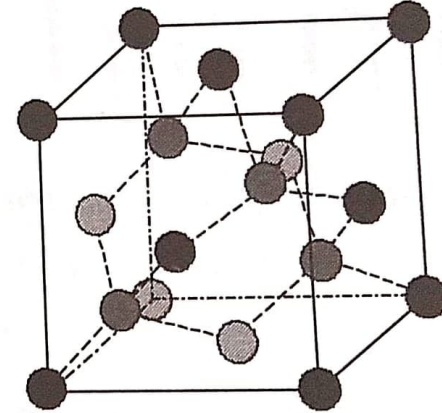
$$C = \frac{2 \times \frac{4}{3} \pi r^3}{a^3} \Rightarrow \boxed{C = \frac{\pi\sqrt{3}}{8} \approx 0,68}$$

Diamant



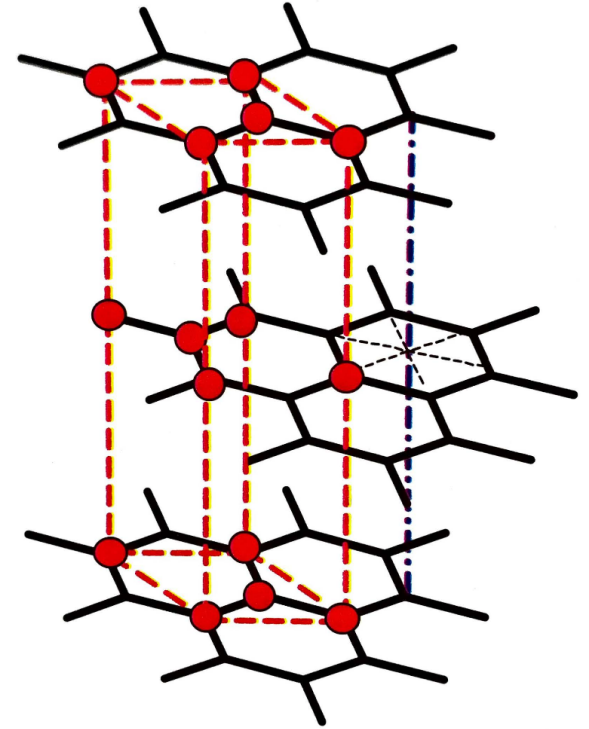
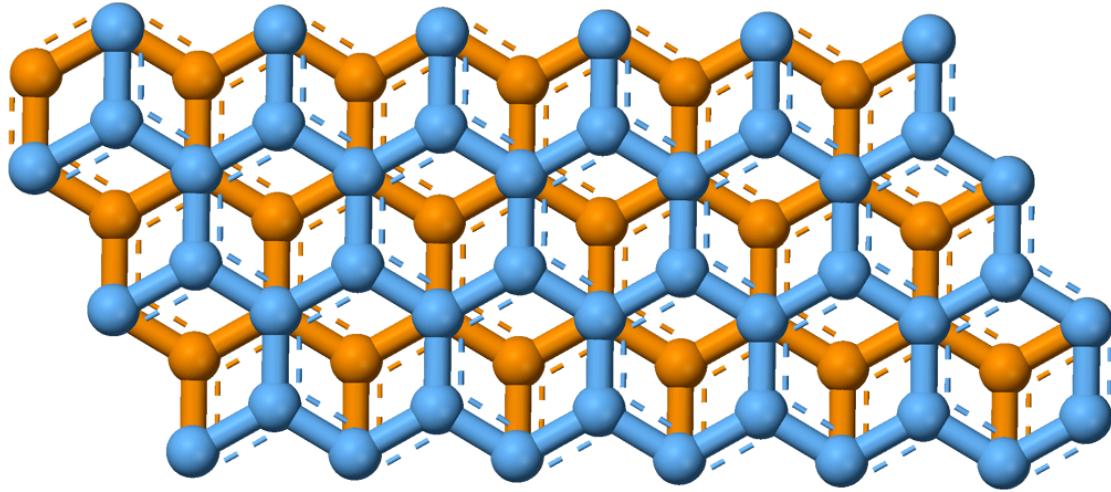
CFC de carbone

4 atomes de C ●
→
dans la moitié des
sites tétraédrique



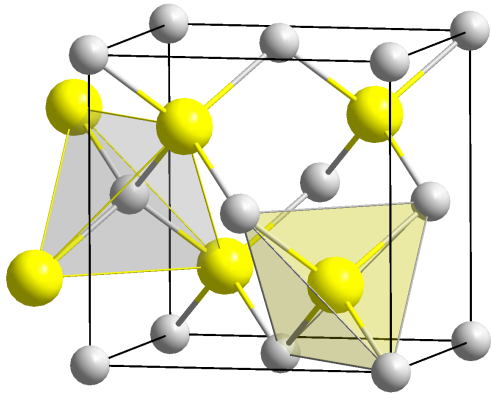
Structure du diamant à partir d'une maille CFC de carbone.

Graphite



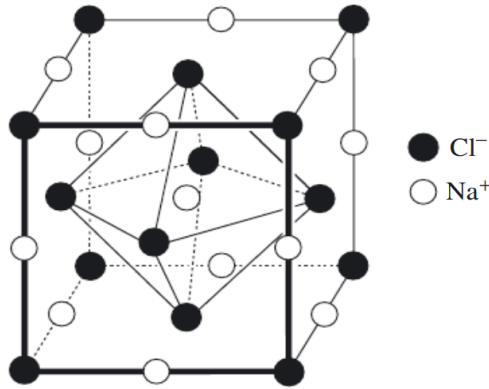
Cristaux ioniques

ZnS



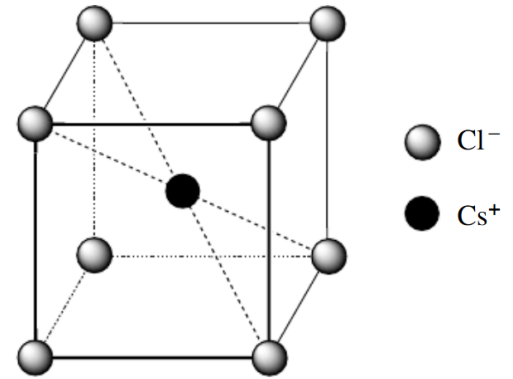
0,40

NaCl



0,54

CsCl



0,80

$\frac{r_+}{r_-}$

Merci